Prospective Attribution of Graph Information and Head Factorization in TSI Transition Models

Chulsu Ha

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Abstract

An earlier 288-cell benchmark could establish exact representability but could not separate graph information from head form because its TSI predictor was the generating equation. We retain that audit and report a new prospective study whose source freeze, fresh 256-bit seed generation, and one-shot execution ran in that order in one scripted sequence. Integrity follows from frozen hashes and the seed commitment, not elapsed delay. The study crosses 30 two-edge graph motifs with three head families, two outside the original linear TSI equation, and reserves stochastic two-coordinate actions that activate both mechanisms simultaneously. Across 120 independent worlds, train-only selection recovered the graph and both head families in every world (multiplicity-adjusted Wilson lower bound 0.941). Correct-minus-wrong graph NLL effects were 0.35933 for the seven-parameter factorized head, 0.36160 for a seven-parameter generic sparse head, and 0.03893 for a 55-parameter dense modular head; all simultaneous lower bounds were positive. With the correct graph, factorized and generic seven-parameter heads were exactly predictive-equivalent. A seven-sparse embedding makes this tie possible, and the sparse procedure recovered the equivalent support in every world. This cohort, its root seed, and its eight-quantity multiplicity family are shared with Paper 3; these overlapping endpoints belong to one frozen analysis, not independent evidence. We also prove population motif identifiability under full-support primitive interventions and symmetric coordinate corruption $p < 6/7$, with a finite repeated-input recovery bound. A post-review 3×3 noise sweep retained 1.000 graph–head recovery in every cell. Outside-family performance was noninferior to the original linear family. In a second frozen cohort, an unmodeled nonadditive synergy destroyed exactness in 120/120 worlds while preserving a learned-versus-wrong graph NLL effect of 0.18545 [0.18388, 0.18702]. A clean-room Node.js implementation reproduced 120/120 learned fits and 10/10 analysis means. The identified contribution is graph information plus recovery of a minimal typed support, not a unique predictive advantage of factorized notation over a generic head that recovers the same support.

1 Introduction

TSI proposes that an internal transition model should represent declared structural dependencies rather than treat all state coordinates as unrelated numbers. The historical standalone studies in Paper 3 evaluate sealed structural OOD, open-loop rollout, and a same-task criterion association in a different simulator. They do not establish the exact factorization studied here, estimate the quantities in the historical Paper 4 audit, or constitute replications. The new prospective extension in Paper 3, however, is this paper's attribution cohort: the root seed and multiplicity family are shared, and overlapping endpoints are not independent evidence.

The original question for this benchmark was whether a discovered structural factorization predicts held-out interventions better than diagonal or unstructured controls. That wording is too strong. The finite generator is inside the TSI hypothesis class, and the fitted TSI predictor uses the same transition equation. Once the graph and six parameters are uniquely recovered, exact prediction follows analytically. The primary audit therefore concerns exact representability and finite-family identifiability, not an empirical causal effect of factorization.

Structural prediction requires a chain from raw observation through entity discovery, cross-time identity, relations, structural variables, and a dependency graph to prediction. This study addresses the later stages of that chain. The entity set and its cross-time identity are supplied by the generator; what we test is graph recovery and prediction once those are given. Discovering entities and maintaining their identity from raw observation are outside the estimand of this audit.

We retain the frozen comparison as a transparent record and add a clearly labelled post-review diagnostic. That diagnostic asks three narrower questions: how cell-level averages conceal ties and structural zeroes; whether graph choice has the same effect under factorized and untied dense heads; and where exactness fails under genuine two-coordinate actions or a transition term outside the fitted TSI family. These diagnostics expose attribution boundaries rather than rehabilitate a universal model-superiority claim. A subsequent prospective study then resolves the identified design defects with a fresh stochastic family, a complete graph-by-head design, matched seven-parameter controls, learned routing, genuine two-mechanism compositions, multiplicity-adjusted world-level inference, and a second-language implementation.

2 Related Work and Scope

Causal structure learning from interventions. Observational DAG discovery is generally limited by Markov equivalence, while known interventions can refine the equivalence class and, under additional conditions, identify more of the graph [8, 19]. Score-based and continuous methods range from interventional greedy equivalence search to differentiable optimization with interventional likelihoods [6, 20]. Our task is substantially narrower: all five coordinates and intervention targets are observed, the candidate graphs are a finite typed motif family, and the data generator lies in the declared transition language. Thus the 120-world result is evidence for recovery within that finite family, not a general DAG-discovery result and not a claim of causal identification from observational data.

Causal representation learning. Causal representation learning asks when latent causal variables and their relations can be recovered from high-dimensional mixtures [14]. Interventional results establish identifiability under assumptions that vary across perfect, imperfect, unknown-target, uncoupled, and multi-node interventions [1, 5, 16, 17]. In contrast, this study does not infer latent coordinates or a perceptual mixing map: the modular coordinates, action target, and action magnitude are supplied. Its contribution is a controlled attribution of graph choice and typed head factorization after those representational variables have already been declared.

Sparse support recovery. Exact support recovery by ℓ_1 methods and greedy pursuit is known to depend on design, signal, noise, and incoherence or restricted-curvature conditions [2, 7, 15, 18]. The seven-sparse embedding used here is an algebraic representation over $\mathbb{Z}_7$, rather than a linear-Gaussian Lasso or orthogonal-matching-pursuit problem. Our population proposition identifies the finite motif under full-support primitive interventions, and its repeated-input corollary controls coordinatewise modal errors. Neither result proves finite random-design recovery for the implemented greedy selector; its 120/120 recovery rate therefore remains empirical.

Structured world models. Interaction networks and graph networks encode object and relation structure as an architectural inductive bias [3, 4]. Neural relational inference, contrastive structured world models, slot-based object discovery, and structured exploration learn graphs or object-centric dynamics from observations of varying complexity [9–11, 13]. TSI's finite modular world is a test bed for a more specific question: whether declared dependency information and transition-head factorization make separately measurable contributions under controlled OOD interventions. The present evidence does not establish visual object discovery, scalable graph-neural dynamics, or real-world validity.

Modularity and independent mechanisms. The independent-causal-mechanisms principle motivates decomposing a data generating process into autonomous modules, and prior work has sought to learn such mechanisms from transformed domains [12]. Our graph-by-head factorial design operationalizes only a restricted analogue: it tests whether routing information still contributes when the prediction head is correctly factorized, untied and dense, or deliberately mismatched. Consequently, the reported interaction and stress-cohort results delimit component attribution inside the benchmark; they do not demonstrate universal autonomy or discovery of independent mechanisms.

3 Audit Contract and Methods

3.1 Finite family and analytic representability

The family has five finite state coordinates with cardinalities $(4, 3, 5, 3, 4)$, three directed dependency graphs, and 96 multiplier/coupling mechanisms, giving 288 graph–mechanism cells. For graph edge $s \to r$, the generator is

$$x'_j = \left[x_j + m_j a_j + \mathbf{1}\{j = r\}\, c\, a_s(1 + x_r)\right]_{q_j}.$$

The fitted TSI head applies this same equation after recovering the graph, $(m_1, \ldots, m_5)$, and c from training transitions. Candidate graphs are scored only on the training partition, and a graph is accepted only when it is the unique candidate with training exact accuracy 1.000. Exact test prediction is therefore guaranteed whenever recovery succeeds; accuracy 1.000 is not a stochastic discovery after conditioning on recovery.

Training actions are the five magnitude-1 one-hot vectors. The original test partition adds held-out primitive source–action pairs and exactly two single-coordinate magnitude-2 actions, $(0, 2, 0, 0, 0)$ and $(0, 0, 2, 0, 0)$. These are scale extrapolations, not two-coordinate compositions. Across all 720 states they produce 1,440 intervention cases per cell. Train/test source–action keys have zero overlap; the recorded split audit, train-only discovery audit, test-only evaluation audit, and source inspection checks all pass.

3.2 Frozen controls and units

The diagonal model estimates coordinate-wise action multipliers. The graph-unaware dense polynomial model uses an intercept, five state values, five action values, and 25 source–action interactions, fitted by least squares. The lookup model memorizes training source–action pairs. The wrong-routed model applies the factorized equation under a shifted graph. The independent unit is the graph–mechanism cell. Five row-permutation seeds are nested within the trainable controls and are averaged within cell; TSI, wrong-routed, and lookup have one deterministic fit. The original descriptive contrast is TSI minus graph-unaware dense intervention accuracy.

3.3 Post-review diagnostics

The following analyses were designed after peer review and are not preregistered confirmatory tests.

First, a graph-by-head panel fills four operational cells: correct versus shifted graph crossed with the modular factorized head versus an untied multi-output least-squares head. The dense head uses an intercept, all five action values, and the graph-selected term $a_s(1+x_r)$; its output coefficients are not tied to the generator parameters. This is one explicit dense-head operationalization, not a universal class of dense models.

Second, a genuine composition panel tests all $\binom{5}{2} = 10$ binary actions having two distinct active coordinates, or 7,200 cases per cell. Third, a misspecified panel tests all five single-coordinate magnitude-2 actions after adding the unmodeled target term

$$a_s(a_s-1)(1+x_r).$$

This term is zero for every magnitude-1 training action, so the original training partition and recovery procedure are unchanged, but it is outside the fitted action-linear TSI equation at test time. The panel contains 3,600 cases per cell.

The frozen audit is stored in `experiments/paper4_final_comparative_audit.json`; the extension is in `experiments/paper4_postreview_diagnostics.json`. Corrected cell distributions and budget/replay declarations are stored in `paper4_statistical_analysis_r2.json` and `paper4_fairness_audit_r2.json`.

3.4 Prospective attribution contract

After the diagnostic, a new contract and implementation were frozen before a fresh confirmatory seed was generated. The freeze, seed generation, and one-shot execution ran in that order in one scripted sequence; the ordering guarantee follows from frozen source hashes and the seed commitment, not elapsed time. A world has five coordinates modulo 7 and a graph motif with two distinct sources entering one target. There are 30 motifs. Each edge uses one of three typed functions,

$$\phi_L = a_s(1+x_t), \quad \phi_Q = a_s(1+x_t)^2, \quad \phi_S = a_s(1+x_s+x_t).$$

The last two are outside the original linear equation. Training and selection contain 800 and 400 disjoint noisy primitive-action transitions. The 1,200 OOD cases contain two active coordinates; one prespecified stratum activates both graph sources and therefore composes two mechanisms. Coordinate noise increases from 0.08 to 0.12 at OOD evaluation.

The complete attribution panel crosses correct versus shifted graph with three head operationalizations. The factorized head has five direct multipliers and two typed edge coefficients. A generic sparse modular head greedily selects seven output–feature coefficients from a dictionary containing five direct actions and all three functions for each declared edge. A dense sensitivity fits all $5 \times 11 = 55$ modular coefficients by coordinate descent. Thus the factorized and sparse-generic cells match active parameter count at seven; the dense cells intentionally give the generic head greater capacity.

Proposition 1 (Seven-sparse embedding)**.** *For a fixed graph and selected edge families, every seven-parameter factorized transition in the prospective family has an exactly equivalent seven-sparse generic modular head.*

Proof. For each coordinate j, assign the generic coefficient on direct feature a_j at output j to m_j, giving five nonzero coefficients. At the common target t, assign the coefficient on selected feature ϕ_{f_e} for edge e to c_e, giving two more. Set every other coefficient to zero. The generic increment is then

$$\delta_j = m_j a_j + \mathbf{1}\{j = t\} \sum_{e=1}^{2} c_e \phi_{f_e},$$

which is the factorized increment coordinate by coordinate. Adding x_j and reducing modulo 7 preserves equality. Conversely, restricting generic support to these seven positions yields the same factorized equation. □

Proposition 2 (Population motif identifiability)**.** *Consider the prospective family over $\mathbb{Z}_7^5$. Suppose the primitive intervention distribution has full support over every state x and unit action e_s, all multipliers and edge coefficients lie in $\{1, 2, 3\}$, and each successor coordinate is independently retained with probability $1 - p$ or replaced uniformly by one of its other six values with probability $p < 6/7$. Then the joint population distribution of (x, e_s, x') uniquely identifies the common-target two-source motif, the ordered head family attached to each source, and all seven parameters.*

Proof. Conditional on an input and a deterministic center value, the center has probability $1 - p$, whereas each other value has probability $p/6$. The inequality $p < 6/7$ is equivalent to $1 - p > p/6$, so the center is the unique conditional mode. The population distribution therefore identifies the complete deterministic primitive-action transition map.

For action e_j, output coordinate j changes by m_j; an edge sourced at j changes only its distinct target coordinate. Hence the five direct multipliers are identified. After subtracting these direct increments, there exists a state at which a primitive action at a nontarget coordinate produces a nonzero residual at another coordinate if and only if that action coordinate is a graph source. Indeed, each declared edge feature is nonzero for some state and each coefficient is nonzero. Full state support therefore identifies the unique output receiving two such residual families and its two sources.

It remains to identify each source's family and coefficient. The source–target feature $\phi_S = 1 + x_s + x_t$ depends on x_s, whereas $\phi_L = 1 + x_t$ and $\phi_Q = (1 + x_t)^2$ do not, so ϕ_S cannot equal a nonzero multiple of either alternative on full support. If $c\phi_L = c'\phi_Q$ for every x_t, evaluation at $x_t = 0$ gives $c = c'$; evaluation at $x_t = 1$ then gives $2c = 4c$ in $\mathbb{Z}_7$, hence $c = 0$, contradicting $c \in \{1, 2, 3\}$. Thus the family is unique. At $x_s = x_t = 0$, every selected feature equals one, so its residual identifies the corresponding coefficient. All components are therefore unique. □

Corollary 1 (Repeated-input recovery bound)**.** *Let $\mathcal{I}$ be any finite separating set of primitive input–output coordinate pairs sufficient for the preceding proof, and suppose each pair's input is repeated independently at least n times. Modal reconstruction on $\mathcal{I}$, and hence motif identification, fails with probability at most*

$$6|\mathcal{I}| \exp\left[-\frac{n}{2}\left(1 - \frac{7p}{6}\right)^2\right].$$

Proof. For a fixed input, center value, and one wrong value, subtract the wrong-value indicator from the center indicator. This variable lies in $[-1, 1]$ and has expectation $(1 - p) - p/6 = 1 - 7p/6 > 0$. Hoeffding's inequality bounds the probability that the wrong empirical count reaches the center count by $\exp[-n(1 - 7p/6)^2/2]$. A union bound over six wrong values and all pairs in $\mathcal{I}$ gives the displayed result. When every modal center is correct, Proposition 2 supplies uniqueness. □

This result is a population/full-support statement and a repeated-input bound; it does not prove that the finite random-design greedy selector must recover the motif. Its 120/120 recovery rate remains an empirical result.

The independent unit is world. The confirmatory cohort has 120 worlds and uses Bonferroni simultaneous intervals for eight inferential quantities at familywise level 0.05. The primary graph effects are wrong-minus-correct NLL within each head.

The analysis object reports ten interval-bearing quantities at the frozen Bonferroni divisor of eight. They contain one exact duplicate pair, one sign-flipped pair, and one deterministic zero, leaving seven informationally distinct stochastic quantities. The frozen v1 contract specified only the integer divisor and did not enumerate its members; this is a reporting defect, not a preregistered family definition. The current code labels the eight gate quantities as identification, learned-routing NLL, factorized-graph NLL, generic-graph NLL, dense-graph NLL, rollout Hamming, criterion Brier, and outside-family noninferiority, explicitly as a post-freeze clarification. A post-review sensitivity using divisor ten leaves every decision unchanged. This cohort, its root seed, and the multiplicity family are shared with Paper 3; overlapping endpoints in the two papers are subsets of one frozen analysis, not independent evidence.

No power analysis for 120 worlds was frozen before confirmation. A post-review retrospective design justification uses only the 24 development worlds, stratified by original versus outside-linear family. Across 20,000 bootstrap draws, estimated conjunctive power for all nine gates at 120 worlds is 0.9694 with Monte Carlo interval [0.9670, 0.9718]. It treats the development-fitted calibration as fixed and inherits the empirical support of 24 worlds. It is neither a preregistration record nor a retrospective change to the frozen design.

3.5 Prospective outside-model-family stress contract

A second source freeze precedes a separate 120-world seed. Its held-out successor adds

$$d\, a_{s_1} a_{s_2} (1 + x_u)$$

at the graph target, where u is not the target or either source and $d \in \{1, 2, 3\}$. This nonadditive synergy is zero on every primitive training and selection action and is absent from all fitted dictionaries. It activates only when both mechanisms are intervened on together. The frozen gates require nonexact learned prediction in every world and a wrong-minus-learned NLL mean of at least 0.03 with a positive 95% lower bound.

4 Results

4.1 Frozen exact-family audit

Table 1 reports the complete 288-cell audit. TSI is exact because the recovered predictor is the generating equation, not because the test identifies a general learning advantage.

The averages conceal important cell structure. Dense intervention accuracy is exact in 33 cells and zero in 32; accordingly, not all 288 TSI-minus-dense cell differences are positive. Wrong-routed accuracy is exactly 0 in 192 cells and 0.6 in 96 cells, while lookup is 0 in all 288 cells. Wrong routing therefore shows graph-dependent support points; it is not a generic test that sparsity alone is insufficient. Lookup is a deliberate input-disjoint structural zero, not a competitive representation learner.

After averaging the five repeated trainable rows within cell, the finite-panel mean TSI-minus-dense intervention contrast is 0.587. A paired cell bootstrap gives $[0.549, 0.624]$; the corresponding

Table 1: Frozen finite-family audit.

Model	Exact accuracy	Intervention accuracy
TSI graph-discovered factorized	1.000	1.000
Dense polynomial trainable	0.622	0.413
Diagonal trainable	0.605	0.410
Wrong-routed factorized	0.234	0.200
Unstructured lookup	0.000	0.000

all-case contrast is 0.378 with $[0.354, 0.401]$. These intervals summarize variation among the enumerated cells. They neither define a superpopulation nor remove the analytic exact-representability confound.

4.2 Post-review diagnostic panels

Table 2 reports mean cell accuracy. All entries use models fit only on the original primitive-action training partition.

Table 2: Post-review descriptive diagnostics. "Original" is the two single-coordinate magnitude-2 actions; "Composition" uses ten two-coordinate binary actions; "Misspecified" adds unmodeled action curvature.

Model	Original	Composition	Misspecified
Correct graph, factorized head	1.000	1.000	0.869
Wrong graph, factorized head	0.200	0.227	0.413
Correct graph, dense head	0.268	0.285	0.231
Wrong graph, dense head	0.501	0.463	0.485
Graph-unaware dense	0.413	0.426	0.392

The genuine two-coordinate panel does not provide an independent discovery: the generator and TSI head are both action-linear, so exact superposition is analytically implied. It does correct the earlier intervention description. Under the unmodeled curvature term, TSI accuracy falls to 0.868889 and none of the 288 cells remains exact. Thus exactness is not robust outside the declared hypothesis family. The post-review TSI-minus-graph-unaware-dense difference is positive in that particular panel, but it was not preregistered and does not by itself identify a universal advantage.

The graph-by-head panel also rejects a simple attribution. On the original actions, correct minus wrong graph accuracy is 0.800 with the factorized head but -0.233 with the specified dense head. Factorized minus dense-head accuracy is 0.732 with the correct graph and -0.301 with the wrong graph. The reversal shows a graph–head interaction tied to these operational definitions; there is no head-independent graph effect and no graph-independent factorization effect. This diagnostic reversal is specific to that least-squares head, exact-family generator, and accuracy endpoint; under the prospective stochastic family and modular dense head, the graph NLL effect is positive.

4.3 Prospective attribution results

Train-only selection recovered the motif and both head families in 120/120 worlds; the multiplicity-adjusted Wilson lower bound was 0.941. Table 3 gives the complete graph-by-head results.

Table 3: Wrong-minus-correct graph composition NLL. Positive values favor the correct graph.

Head	Parameters	Mean	Simultaneous interval
Factorized typed	7	0.35933	[0.35431, 0.36435]
Generic sparse	7	0.36160	[0.35683, 0.36637]
Generic dense	55	0.03893	[0.02527, 0.05259]

Graph information has a positive effect under all three head operationalizations, including the higher-capacity dense sensitivity. With the correct graph, factorized minus generic-sparse NLL is exactly 0.00000 in every world. The graph-by-head interaction for the two seven-parameter heads is -0.00227 with simultaneous interval [-0.00822, 0.00368]. The proposition explains the exact predictive tie: both fitted procedures recover the same seven-term function. Hence the evidence attributes performance to correct graph information and minimal-support recovery, not to an irreducible advantage of factorized notation.

The equivalence is conditional. It requires that the generic dictionary contain the selected edge functions and that the selector recover that support. Where the dictionary is misspecified, or where the support is not recoverable at the given sample size and noise level, the typed factorization is not equivalent to the generic head. That regime is not tested here, and it is the natural place to look for a distinct contribution of typed structure.

The 80 worlds using at least one nonlinear head were noninferior to the 40 original linear worlds. Their NLL difference was -0.00132 with simultaneous interval [-0.01031, 0.00766], below the frozen margin 0.03. All nine attribution and linked Paper 3 gates passed.

A separate post-review descriptive sensitivity crossed train noise $\{0.04, 0.08, 0.16\}$ with OOD noise $\{0.06, 0.12, 0.24\}$, using 120 public-seed worlds in each of nine cells. Graph and head recovery was 1.000 in every cell. Across the grid, the minimum mean wrong-minus-correct NLL effects were 0.23337 for factorized, 0.23773 for generic sparse, and 0.02420 for generic dense heads; the smallest dense-head descriptive 95% lower bound was 0.01719. The panel was not confirmatory and varies only coordinate-corruption noise.

4.4 Outside-model-family stress results

The nonadditive synergy destroyed exact learned prediction in all 120 fresh worlds. Mean noiseless center accuracy in the both-mechanisms stratum was 0.14104. Nevertheless, wrong-minus-learned stochastic composition NLL was 0.18545, with world SD 0.00879 and 95% interval [0.18388, 0.18702]. Thus the graph effect remains after exact representability has been deliberately removed. The synergy was zero in every training and selection case, and all four frozen stress gates passed.

A Node.js implementation with no imports from the Python project independently recovered the learned model and NLL values for all 120 attribution worlds and recomputed all ten published analysis means. It also verified the revealed root-seed commitment; no discrepancy was found.

A separate Python audit derived all 120 world seeds from the revealed root and regenerated every exported train, selection, and test transition with the archived frozen generator. All cases matched the portable input. This closes seed-derivation and export consistency within the frozen Python path, but it does not turn the Node.js analysis into an independent generator implementation.

4.5 What this audit does and does not establish

Table 4: Scope of the established claims.

Established	Not established
Train-only graph and head recovery in 120/120 worlds	Any advantage of factorized *notation* over a generic head that recovers the same support
Correct graph improves held-out two-mechanism composition NLL under all three head operationalizations	Entity discovery, or identity maintenance from raw observation
The graph effect survives deliberate removal of exact representability (0.18545, 120/120 worlds nonexact)	That the richer structural state adds value beyond a correct minimal support
Factorized and matched generic sparse heads are exactly equivalent once the same support is recovered	Real-world, neural, continuous, or perceptual validity

5 Scope

The historical exact-family and retrospective diagnostic results retain their original interpretation and are not counted as prospective evidence. The two new cohorts use independent fresh seeds, source freezes, one-shot locks, and world-level inference.

The prospective population is finite and synthetic: five modular coordinates, 30 two-edge motifs, three declared additive head families, one unmodeled synergy form, and specified coordinate noise. The conclusions quantify this population. They do not imply universal superiority of sparse models or a claim about visual, continuous, neural, or real-world systems, because those objects are not sampled by the estimand.

The seven-parameter tie is a positive structural result, not a failed comparison. The proposition shows that factorized and generic heads are functionally identical once the generic procedure recovers the same support. The empirical distinction is support and graph recovery. The 55-parameter dense sensitivity retains a smaller but positive graph effect, demonstrating that the result is not created solely by matched sparsity.

The clean-room Node.js path independently implements graph/head search, factorized fitting, NLL evaluation, and analysis means without importing the Python project. It is an implementation replication by the same investigator, not evidence about between-laboratory variation. This authorship fact does not alter the computational reproducibility claim made here. The path consumes exported world data and does not re-derive worlds from the committed seed; generator-side errors and world-seed derivation are outside its coverage. The separate Python derivation audit verifies the declared seed rule and complete export against the archived frozen generator, but remains within the same language and code lineage. No independent laboratory replication is claimed.

6 Conclusion

The original 288-cell audit established exact representability but confounded the generating equation, graph information, and head form. The retrospective diagnostic exposed that defect; it did not solve it.

Two newly frozen prospective cohorts do. In the attribution cohort, train-only selection recovered two-edge graphs and head families under stochastic noise. Correct graph information improved held-out two-mechanism composition NLL under factorized, parameter-matched generic sparse, and larger dense heads. The seven-sparse embedding proposition establishes an equivalent generic support, and the sparse procedure recovered that support in every world, producing an exact tie with the factorized head. The distinctive object is therefore the typed minimal support and its learned graph, not surface notation.

The population result now identifies that motif and its typed heads under full-support primitive interventions and symmetric corruption below 6/7; the finite bound states the repeated-input coverage required for modal recovery. The random-design greedy selector is not covered by that theorem, but its 120/120 recovery is stable across the post-review 3×3 noise grid.

In the separate stress cohort, an unmodeled nonadditive synergy made the fitted model nonexact in every world, yet the learned graph retained an NLL effect of 0.18545 over wrong routing. This removes exact generator–predictor equality as an explanation of the graph effect. A clean-room second-language implementation reproduced every learned fit and analysis mean in the main cohort. Within the stated finite stochastic population, graph information, minimal structural support, composition, misspecification, and post-export fitting/evaluation/analysis implementation reproducibility are therefore separately identified.

The next decisive comparison is not a factorized head versus a generic sparse head. It contrasts two models that have both recovered the same dependency graph under entity creation and deletion, identity maintenance, and relation reconfiguration. For that comparison to identify anything, both arms must receive identical observable information and differ only in representational commitment. Supplying identity or typed relations as input to one arm only would make the result an artifact of information asymmetry, the same class of confound diagnosed in the original finite-family audit. The turnover quantity $b(p)$, its zero criterion, and the vacuity proposition for survivor-only preservation are already formalized in the companion theory; a turnover-focused study is the smallest self-contained instance of this design.

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